

First sprint plan for **Biodex**

Sprint duration: 2 weeks - week 5-7

Sprint goal: to create a functional prototype (medium fidelity) through multiple iterations

Sprint start	16/08/2026
Sprint end	6/09/2026

Sprint schedule:

Time	Main Activity	Team Activity
Day 1	Planning & task delegation	Agree and decide on UI, git and task delegation
Day: 2–3	Core development	Discuss what to implement for low fidelity, make the SQL database.
Days 4–5	brainstorming	Bernard will tidy up the user stories and make user stories.
Days 6–7	Low fidelity design	Design the low fidelity with basic buttons, text box and basic UI and design.
Day 8	feedback	
Day 10-12	Final development	Each person responsible for their pages go through low fidelity and make sure their idea is coherent with the prototype
Day 12-14	Refinement	Each member goes through the medium fidelity and provide feedback and apply updates

1. Project & Application Foundation

Responsible: Team effort

The first checkpoint is to build the foundational piece for everyone to work on, such as the SQL structure, JFX project structure. The Team should come up with a coherent design philosophy and UI, colour schemes. Should come to an agreement on titles, naming

convention, dimensions, navigation bar, animations, UI components before everyone starts developing their individual sections. This is crucial to make the app look coherent.

Tasks:

- Set up the JavaFX project.
- Configure SQLite/JDBC.
- Establish Git/repository structure.
- Create shared navigation/scene routing.
- Establish basic application styling.
- Create the initial database structure.
- Agree on shared components such as buttons, forms, cards and error messages.

Sprint output: A runnable application with the basic project structure and navigation framework.

2. Login & Signup

Responsible: Ali

Ali will develop the Login/Signup specification, outline specification requirements, and decide which specs to include. His task is to identify Ali as the page owner develops and sets the specs to show what the login and sign up process becomes.

Medium fidelity implementation should include the Login and Signup screens, form validation, and seamless navigation between them. Successful authentication should create the application session because other areas such as settings and upload rely on knowing which user is currently signed in.

Tasks:

- Build a Login screen.
- Build Signup screen.
- Add username/email and password fields.
- Implement basic validation.
- Implement login/signup buttons.
- Connect authentication to the database.
- Create the user session.
- Add navigation to Forgot Password and 2FA.
- Implement logout.

Dependency: Important because vinny will take on Alis design and develop the medium fidelity design, make sure everything is coherent and identify who submitted a sighting.

Sprint output: A user sign up,, log in and reach the main application.

3. Forgot Password & 2FA

Responsible: Forgot Password: Vinny; 2FA: Tom

Support: Ali for Login integration

To be developed in the first three days, finalise settings and requirements. These features can be developed along with the login/sign up because they are all similar flows. The team members should communicate so the flow is consistent across the three screens. 2FA implementation might be difficult but Tom will develop the 2FA verification screen. It will have a six digit email code, expiry handling, resend and button back up in case authentication fails.

Tasks:

- Create Forgot Password screens.
- Separate group meeting
- Create verification-code interface.
- Create a reset-password interface.
- Add navigation between the three stages.
- Create a 2FA six-digit verification screen.
- Add expiry/error states.
- Add resend functionality.
- Create the flow so the three screens are connected.

Dependency: Ali needs to provide the Login navigation points, while Vinny+Tom need to maintain the shared password/verification rules.

Sprint output: A user can access password recovery or 2FA from the Login flow and return to the application.

4. Species Search & Species Details

Responsible: Vinny

Covers the main search functionality that allows the user to find invasive species, while Species Details displays information about the selected species and its local sightings. This page should contain detailed information on the species but it can be refined further.

Species Details should be developed into two connected panes - a Species Detail pane and a Local Sightings pane.

Tasks:

- Build Species Search screen.
- Add search/filter interface.
- Display species results.
- Create Species Details screen.
- Display species image and information.
- Add a threat profile.
- Display local sightings.

- Add "Log Sighting".
- Add "Add Photo".
- Add "Open Full Map".

Sprint output: A user can search for a species, open its details and move into reporting/map functionality.

5. Upload / Report Sighting

Responsible: Dong Hyeon

The Upload section is where users can upload individual species and update sightings, severity or provide feedback/fix mistakes. The specification requires users to upload an image and provide information such as location, species, date, notes and severity. Some fields of entry will be optional, auto filled or required. Admin should have an individual page to control and moderate but it will not be implemented in this fidelity design.

Tasks:

- Create an upload/report screen.
- Add file picker.
- Add image preview.
- Add species field.
- Add location field.
- Add date and notes.
- Add severity selection.
- Validate required information.
- Add submission confirmation.
- Store the basic sighting information.
- Display submission status.

Dependency: The page can only be interacted after the user is logged in and must be associated with the current user. Anonymous users may interact with the app but only verified and vouched for users can write and contribute.

Sprint output: A signed-in user can create and submit a basic sighting report.

6. Heat Map

Responsible: Joshua

The heatmap section adds visual feedback of various sightings across Brisbane. It should be Brisbane centred, with various filters for time and species and a button for the report signing section that will take users to the upload page. The scope might exceed the scope for this checkpoint so the team should get a basic map working and make sure it's not crowded.

Tasks:

- Create a map screen.
- Centre map on Brisbane.
- Display sample/initial sighting locations with emphasis on simplicity
- Add basic species filter.
- Add basic date filter.
- Add map legend.
- Add sighting interaction.
- Connect selected sightings to Species Details.
- Add Report Sighting button.

Dependency: The Heat Map becomes more useful once the Upload functionality is producing sighting records. Therefore, the first sprint can use sample data while the database integration is being completed.

Sprint output: A working map interface showing invasive-species sightings and basic filtering.

7. User profile

Responsible: Yash

The Settings page is a sub-view of the **Profile Page**, accessible by side-navigation item alongside **My Uploads** and **My Photos**. The team should implement an easy to use user profile with controls to the account through photo/upload behaviour, pest-identification behaviour, notifications, and privacy/data controls. All settings are user-scoped, persisted in a relational database, and loaded once per session into memory for fast UI binding." action.

For Sprint 1, the team should focus on getting the buttons in place, coherent and capable

Tasks:

- Add radio buttons for notification and privacy.
- Add profile
- Add customisation (light/dark mode).
- Add my uploads check
- View and edit account details
-

Dependency: require the login to be completed beforehand, will sit alongside user upload, view threats. Will not be accessible for non logged in users.

Sprint output: a coherent design philosophy that doesn't bog down users. Light/dark mode does not have to be implemented immediately nor does the display of the user profile. Main focus should be on the flow and UI, compactness.

Sprint completion:

The sprint plan can be concluded when the team has a successful medium fidelity of Biodex with varying features as outlined below.

Satisfactory:

Working clickable navigation between all major screens
Search/filter interaction
Species cards/results
Detailed species information
Threat-level indicators
Local sightings
Map with realistic sample sightings
Upload screen with image preview
Report form with validation states
Capable flow for login, forgot and reset password
Basic profile/settings screens
Confirmation after submission
2FA screen as a prototype interaction

Minimum requirements:

Area	Minimum Requirement
Navigation	Good design with transition between pages
Login	Login screen exists with text input but no verification or hashing implemented
Home/Search	Good user design where user can search for species
Report sighting	Users can upload and report sightings

Species details	Users can view information about species
Map	Users can view heatmaps
Consistent UI	Coherent design philosophy throughout
Main interaction	Buttons, forms, areas working and properly implemented
Error states	Basic levels of user feedback for forbidden attempts